

Practical - 6

1. Client Server using Named Pipes(FIFO)

server.c

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <string.h>
4#include <fcntl.h>
5#include <unistd.h>
6#include <sys/stat.h>
7#define FIFO1 "client_to_server"
8#define FIFO2 "server_to_client"
9#define SIZE 100
10int main() {
11    char message[SIZE];
12    char response[50];
13    // Create FIFOs
14    mkfifo(FIFO1, 0666);
15    mkfifo(FIFO2, 0666);
16    printf("Server started...\n");
17    while (1) {
18        // Open FIFO for reading
19        int read_fd = open(FIFO1, O_RDONLY);
20        // Read client message
21        read(read_fd, message, SIZE);
22        close(read_fd);
23        printf("Client: %s\n", message);
24        // Exit condition
25        if (strcmp(message, "exit") == 0)
26            break;
27        // Process message
28        sprintf(response, "Server received: %s", message);
29        // Open FIFO for writing
30        int write_fd = open(FIFO2, O_WRONLY);
31        write(write_fd, response, strlen(response) + 1);
32        close(write_fd);
33        printf("Response sent.\n");
34    }
35    unlink(FIFO1);
36    unlink(FIFO2);
37    printf("Server stopped.\n");
38    return 0;
39}
```

client.c

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <string.h>
4#include <fcntl.h>
5#include <unistd.h>
6
7#define FIFO1 "client_to_server"
8#define FIFO2 "server_to_client"
9#define SIZE 100
10
11int main() {
12    char message[SIZE];
13    char response[SIZE];
14    while (1) {
15        printf("Enter message: ");
16        fgets(message, SIZE, stdin);
17        message[strlen(message, "\n")] = '\0';
18        // Send message
19        int write_fd = open(FIFO1, O_WRONLY);
20        write(write_fd, message, strlen(message) + 1);
21        close(write_fd);
22        // Exit
23        if (strcmp(message, "exit") == 0)
24            break;
25        // Receive response
26        int read_fd = open(FIFO2, O_RDONLY);
27        read(read_fd, response, SIZE);
28        close(read_fd);
29        printf("Server: %s\n", response);
30    }
31    return 0;
32}
```

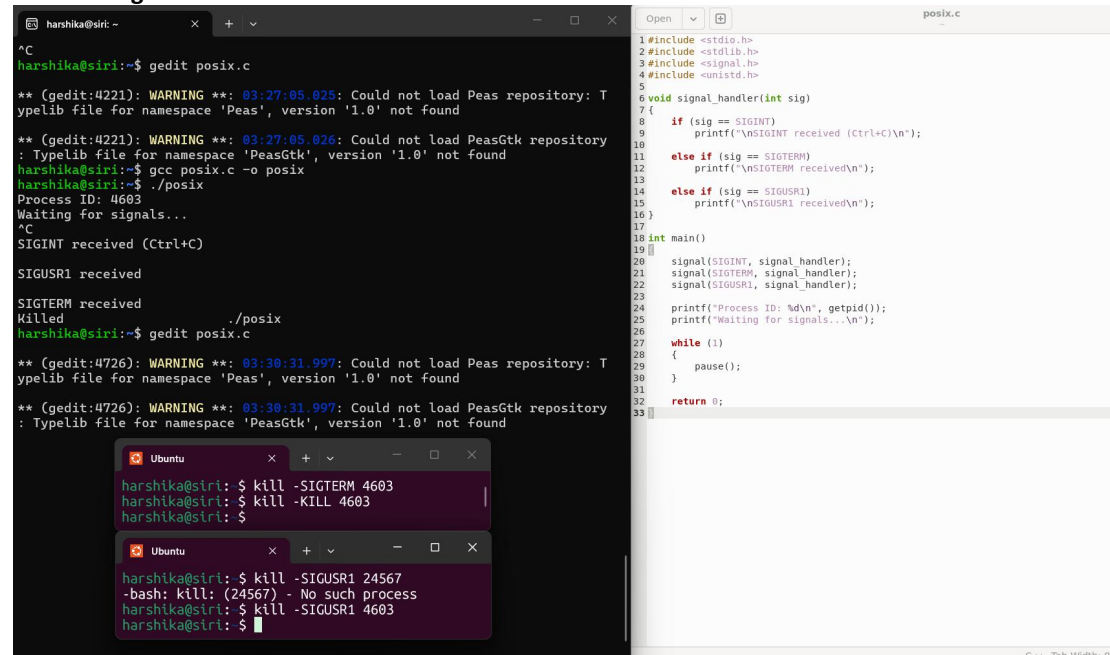
Ubuntu

```
harshika@siri:~$ ./client
Enter message: Hello Server
Server: Server received: Hello Server
Enter message: client 1
Server: Server received: client 1
Enter message: client 2
Server: Server received: client 2
Enter message:
```

harshika@siri:~\$ gedit server.c

```
** (gedit:3960): WARNING **: 03:22:14.102: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:3960): WARNING **: 03:22:14.103: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gedit client.c
** (gedit:3390): WARNING **: 03:22:43.292: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:3390): WARNING **: 03:22:43.292: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
^C
harshika@siri:~$ gedit client.c
** (gedit:3667): WARNING **: 03:23:22.476: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:3667): WARNING **: 03:23:22.476: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc server.c -o server
harshika@siri:~$ gcc client.c -o client
harshika@siri:~$ gedit server.c
** (gedit:4020): WARNING **: 03:23:43.604: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:4020): WARNING **: 03:23:43.604: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
[[Aharshika@siri:~$ gedit server.c^C
harshika@siri:~$ ./server
Server started...
Client: Hello Server
Response sent.
Client: client 1
Response sent.
Client: client 2
Response sent.
```

2. POSIX Signal



The image shows a development environment with a terminal window on the left and a code editor on the right. The terminal window displays the execution of a C program that demonstrates POSIX signals. The code editor shows the source code of the program, `posix.c`.

Terminal Output:

```
harshika@siri:~$ gedit posix.c
^C
** (gedit:4221): WARNING **: 03:27:05.025: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:4221): WARNING **: 03:27:05.026: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc posix.c -o posix
harshika@siri:~$ ./posix
Process ID: 4603
Waiting for signals...
^C
SIGINT received (Ctrl+C)

SIGUSR1 received

SIGTERM received
Killed
harshika@siri:~$ gedit posix.c
** (gedit:4726): WARNING **: 03:30:31.997: Could not load Peas repository: T
ypelib file for namespace 'Peas', version '1.0' not found
** (gedit:4726): WARNING **: 03:30:31.997: Could not load PeasGtk repository
: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

Code Editor Content (posix.c):

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <signal.h>
4 #include <unistd.h>
5
6 void signal_handler(int sig)
7 {
8     if (sig == SIGINT)
9         printf("\nSIGINT received (Ctrl+C)\n");
10
11     else if (sig == SIGTERM)
12         printf("\nSIGTERM received\n");
13
14     else if (sig == SIGUSR1)
15         printf("\nSIGUSR1 received\n");
16 }
17
18 int main()
19 {
20     signal(SIGINT, signal_handler);
21     signal(SIGTERM, signal_handler);
22     signal(SIGUSR1, signal_handler);
23
24     printf("Process ID: %d\n", getpid());
25     printf("Waiting for signals...\n");
26
27     while (1)
28     {
29         pause();
30     }
31
32     return 0;
33 }
```

Terminal Output (kill commands):

```
harshika@siri:~$ kill -SIGTERM 4603
harshika@siri:~$ kill -KILL 4603
harshika@siri:~$
harshika@siri:~$ kill -SIGUSR1 24567
-bash: kill: (24567) - No such process
harshika@siri:~$ kill -SIGUSR1 4603
harshika@siri:~$
```